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An early Spanish vocabulary for children who use AAC: developmental and linguistic considerations

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ABSTRACT

For children with complex communication needs in the early stages of language development, access to appropriate vocabulary provides a means for social interaction and participation, and the foundation for the acquisition of grammar and other language related skills. While numerous resources are available to support decision making for speakers of English, there is a pressing need to rapidly expand such resources for other languages. Spanish is the official language in 20 countries, and in other countries (e.g., United States) Spanish-speaking communities represent a substantial proportion of the population. The aim of this study was to produce a developmentally-relevant word list for use by Spanish-speaking children in the early stages of language development. The list was developed from an analysis of overlap between published and validated lists of words produced by young Spanish speaking children with typical development. The list includes a wide range of word classes and semantic categories and is proposed as a tool to assist professionals, families and software developers in the process of selecting an initial lexicon for children who require AAC and are learning Spanish. Implications of our findings for vocabulary selection and future research directions are discussed.

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As the use of augmentative and alternative communication (AAC) continues to expand in Spanish-speaking countries and other countries with large Spanish speaking communities (e.g., United States), questions arise about the nature and content of vocabulary selection for an initial AAC system in Spanish. The topic of vocabulary selection for an initial lexicon (i.e., the first words to be taught) for children with atypical language development has been the center of considerable debate for several decades (e.g., Fristoe & Lloyd, 1980; Holland, 1975; Lahey & Bloom, 1977; Judge et al., 2020; Laubscher & Light, 2020; Quick, Erickson, & Mccright, 2019). For instance, Holland (1975) advocated for a selection of a relatively small (35 words) "core lexicon in which a significant proportion of the lexical items can function in more than one grammatical slot" (p. 519). Lahey and Bloom (1977) proposed that an initial lexicon should be categorized into substantive words and relational words. Substantive words refer to nouns, while relational words refer to other parts of speech such as attributes, verbs, prepositions, etc. They suggested that substantive words should be child specific and should be chosen to reflect the child's interests as well as the frequency with which a child comes into contact with particular objects, people, animals, etc. However, initial relational words, they argued, should be generic enough, so they could be used frequently and across multiple contexts.

Despite differences about the specific content and organization of an initial lexicon, both Holland (1975) and Lahey and Bloom (1977) agreed that the first words to be taught to a child with a communication disorder should: (a) be developmentally appropriate (i.e., selected among words that children with typical development use first); (b) be relevant to the child's communicative needs outside the clinic and/or classroom; and (c) include different types of word classes to support the development of grammar. While debate continues over vocabulary selection, these early perspectives set down guidance for decision making that continue to inform practice over 40 years later.

For children and other individuals who are in the early stages of language development and not yet able to spell their messages, best practices in the field of AAC indicate that vocabulary selection should be conducted collaboratively by family members and professionals to ensure that the vocabulary is personally meaningful, relevant in the home, culturally appropriate, functional across contexts, supportive of social interaction, and facilitative of language and literacy development (see Bean, Cargill, & Lyle, 2019 for an extensive discussion). To that end, AAC professionals are encouraged to draw on a combination of sources to identify potential vocabulary that can then be discussed with families and other professionals in a shared decision-making process. These include observation of the child with peers and

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siblings in various activities, analysis of the communication demands of different situations, and use of communication diaries and published resources, such as developmental language inventories, curriculum guides, and core vocabulary lists.

In our field, core vocabulary refers to high-frequency words that make up approximately 80% of an individual's expressive vocabulary and are common across activities, contexts, places, topics and demographic groups (Bean et al., 2019; Witkowski & Baker, 2012). Core vocabulary lists are typically created by sampling language use in speakers at different ages (e.g., toddlers, school-aged, adults), across different modalities (e.g., oral, written) and settings (e.g., school, work place) and with different learning profiles (e.g., neurotypical, neurodiverse). The samples are then analyzed to generate word frequency counts and to calculate the total number of words, the total number of different words, and the number of speakers who used each word. The resulting lists typically consist of 200–300 words that include a range of open (e.g., nouns, verbs, adjectives, adverbs) and closed word classes (e.g., pronouns, auxiliary verbs, copula, prepositions, articles).

Core vocabulary is relatively small and shows little variation across people and settings. It includes a range of word classes that allow for novel utterance generation and word combination, and can serve multiple syntactic, semantic, and pragmatic functions. Because it is limited and consistent across contexts, core vocabulary can be displayed in a prominent and stable location on a communication device and its frequent use can lead to automaticity through development of a consistent motor plan (e.g., Banajee, DiCarlo, & Stricklin, 2003; Bean et al., 2019; Deckers, Van Zaalen, Van Balkom, & Verhoeven, 2017; Trembath, Balandin, & Togher, 2007). In addition, the use of core vocabulary can assist adult partners in the provision of consistent and frequent modeling (e.g., aided input) and elicitation opportunities for use of the vocabulary across a variety of contexts, people, and topics. In a recent study, Quick et al. (2019) found a considerable overlap between the 267 words most frequently used by 16 mothers during interactions with their infants and toddlers with typical development and three lists of words most frequently used by 65 toddlers and preschoolers with typical spoken language development. Quick et al. (2019) suggest that the symmetry between maternal child-directed speech and the most frequent words used by young children with typical development can have important clinical implications for vocabulary selection for children with complex communication needs.

Despite the proposed benefits for the use of core vocabulary, a recent study calls to question the appropriateness of using a core word approach when selecting and teaching expressive vocabulary for early symbolic communicators, that is, children who have begun to produce their first words but have not yet started to combine them into two or more word utterances (Laubscher & Light, 2020). Laubscher and Light (2020) compared five core vocabulary lists developed for English speaking young children aged 2–6 years of age (Banajee et al., 2003; Beukelman, Jones, & Rowan, 1989; Fried-Oken & More, 1992; Marvin, Beukelman, Brockhaus, & Kast, 1994; Trembath et al., 2007), with the words included in the English version of the MacArthur Bates Communicative Development Inventories (MBCDIs; Fenson, Marchman, Thal, Dale, & Reznick, 2007). There are two versions of the MBCDI: (i) Words & Gestures (adapted mostly for toddlers 8–18 months of age and assessing both word production and comprehension), and (ii) Words & Sentences (designed for the assessment of word production in older children, 16–30 months of age). The MBCDI has been normed with over 1,000 infants and 1,400 toddlers, and found to be a valid and reliable tool for assessing the early vocabulary skills of children with and without disabilities (see Laubscher & Light, 2020 for extensive references).

Laubscher and Light (2020) found that approximately 80% of the words included in the MBCDI were not captured in most of the core vocabulary lists for young children. They attributed this finding to the fact that the MBCDI includes a large number of nouns and other content words (i.e., open class words), while the core vocabulary lists privilege function words (i.e., closed class words). Also, children who participated in the studies that produced core vocabulary lists had language abilities equivalent to or above those of a 24-month-old child with typical development. These children would be expected to produce multi-word utterances and even grammatically advanced phrases and sentences which would include function words such as pronouns, prepositions, etc. In contrast, the expressive vocabularies of early symbolic communicators tend to be dominated by content words referring to salient and personally meaningful objects, animals, people, attributes, and actions. Laubscher and Light call for a combination of methods to select developmentally appropriate vocabulary for early symbolic communicators learning AAC, including environmental inventories, communication diaries, categorical inventories, and published and validated inventories of words (i.e., word lists) produced by children of similar age such as the MBCDI.

Word lists and vocabulary inventories vary according to language. Lists from one language cannot and should not be assumed to transfer to another language, as words represent the concepts and grammar of each language (e.g., Jackson-Maldonado, Thal, Marchman, Bates, & Gutierrez-Clellen, 1993; Mngomezulu, Tönsing, Dada, & Bokaba, 2019). Cross-linguistic analyses of vocabulary in young children reveal lexical differences across different languages (e.g., Bornstein et al., 2004), and within different variations of the same language (e.g., Díaz, Fonseca, Bohórquez, Guechá, & Sellabona, 2011; Mariscal et al., 2007). A word that is frequently used by children in one language may not be translatable or developmentally appropriate in another. Likewise, due to variations in grammatical structure, lexical items such as copulas, auxiliary verbs, pronouns, etc., which are frequently part of core vocabulary lists in any given language, may be represented differently and be acquired in different order in different language typologies (Bornstein et al., 2004; Łuniewska et al., 2016). Consequently, a number of core vocabulary lists has been published in a number of different languages, including Afrikaans (Hattingh & Tönsing, 2020), English as a second language (e.g., Boenisch & Soto, 2015), French (Robillard,

Mayer-Crittenden, Minor-Corriveau, & Bélanger, 2014), German (Boenisch, Musketa, & Sachse, 2007), Korean (e.g., Shin & Hill, 2016), Mandarin (e.g., Liu & Sloane, 2006), and Zulu (Mngomezulu et al., 2019).

Although a number of studies provide vocabulary lists and inventories for selecting and teaching vocabulary for children of different ages who use AAC and are learning English, limited resources exist in Spanish. There are 20 countries in the world with Spanish as an official language. Additionally, information about vocabulary selection in Spanish is also relevant to a country like the United States where Spanish speakers represent a substantial proportion of children in public schools. For instance, in California during the 2018–2019 school year, 55 percent of public-school students were identified as Hispanic or Latino (California Department of Education, 2019). Given the importance that vocabulary selection has in the AAC field, the growing number of Spanish speaking AAC users, and the proliferation of language communication apps and software in Spanish, it is imperative to begin to develop tools and resources to assist families, professionals and AAC software developers in the selection of Spanish vocabulary.

The purpose of this study was to generate a developmentally appropriate list of words that could be used in combination with other methods to assist families and professionals in the selection of an initial lexicon for children who use AAC and are in the early stages of Spanish language development. In order to compile a list of highly frequent, developmentally appropriate early words in Spanish, we drew on published lists of frequently used vocabulary by young Spanish-speaking children. To ensure that our list included developmentally appropriate words, we followed Laubscher and Light's (2020) procedures and compared those lists with a Spanish version of the MBCDI (Jackson-Maldonado et al., 1993).

Method

Inclusion criteria

Published lists of words produced by children acquiring Spanish as a native language were included in this study if they met the following criteria: (a) were published in a peer-reviewed journal, (b) included vocabulary lists developed by sampling native speakers of Spanish, and (c) included participants that were 4 years of age and younger. Lists were excluded if they were developed from written language samples, included older children or adults, or non-native speakers of Spanish.

Search methods

Five databases (JSTOR, Educational Resources Information Center (ERIC), Linguistics and Language Behavior Abstracts (LLBA), PsychInfo, and PubMed) were searched using the following sets of free-text search terms, in Spanish and English: (a) initial lexicon/léxico inicial, (b) children/niños, (c) core vocabulary/vocabulario de uso frecuente, and (c) Spanish/Español. An initial yield of 239 articles was identified following the initial search of databases. All articles identified in the database searches underwent a title and abstract screening to identify relevance and evaluate against the inclusionary criteria. Following this, studies underwent a full-text review. An ancestral search was performed on relevant articles and authors to identify additional publications. Only two studies met the inclusion criteria (Manzano, Piñeiro, & Pereira, 1997; Piñeiro & Manzano, 2000). The studies used the same participants but reported on different results (Manzano et al., 1997; Piñeiro & Manzano, 2000).

Participants

The studies by Manzano et al. (1997) and Piñeiro and Manzano (2000) reported on the vocabulary of the same set of participants. The participants included 96 boys and 104 girls, ranging in age from 11;16 to 49;16 (months; days). All the children were typically developing and did not have auditory problems. The participants were recruited from six day care centers located in four different municipalities of Havana, Cuba. The socioeconomic status of the families was reported as average based on the general Cuban population and all of the mothers had at least a high school education.

Data collection and analysis procedures

Manzano et al. (1997) used audio recordings of the children as they were interacting with peers, playing with toys (e.g., teddy bears, plastic toys, musical instruments, cars, balls), and talking with adults in an observation room. Each observation lasted about 30 min. A total of four observers transcribed the language samples. A word was excluded if the observers could not agree or did not understand it. About 7% of the words were omitted. Words containing phonological errors were included as long as the word was recognizable in the context (e.g., *apato* instead of *zapato*/shoes). Additionally, word substitutes that are conventional labels for objects and actions in child vocabulary were included and labeled "stable substitutes" (e.g., "*tic-tac*" for *reloj*/watch; "*pun pun*" for *pelota*/ball).

The participants were grouped in age groups by 2–4 month intervals (e.g., 11;16–14;15, 14;16–17;15, and so on). The resulting groups ranged from 16–20 children. The researchers calculated the frequency count of each word and noted the age group where a word was first produced. The resulting database included a total of 15,428 tokens and 1,259 types.

In 2000, Piñeiro and Manzano further analyzed the list compiled by Manzano et al. (1997), and created a database for use in cognitive/developmental psychology and psycholinguistics research. Their database included new variables such as the first time a word was uttered (FTU), defined as the age range in which a word was uttered for the first time by at least one child. They also defined the Age of Acquisition (AoA) as the age range in which a given word's frequency reached 10% of its total frequency. The authors offer *gato* (cat) as an example. The word was first produced

by a child between 14;16 and 17;15 months, making that range its FTU. Yet, the AoA for *gato* is 17;16–20;15 months because children in this age group produced at least 10% of the word's total frequency.

The authors also provide the number of different words produced for the first time in each age range. The greatest number of different words were produced during the interval of 22;16–25;15 months, which the authors attribute to the vocabulary spurt. Additional variables in the database included the number of letters and syllables, grammatical class, stress pattern, semantic category, and orthographic consonant-vowel pattern for each word.

List organization

The word list generated by Manzano et al. (1997) is organized chronologically based on the age range in which a word was first produced. Within a given age group, the words are listed alphabetically. The frequency count of each word is included. In the case of stable substitutes, the adult form is provided in parenthesis [e.g., *gua gua* for *perro*/dog].

The list generated by Piñeiro and Manzano (2000) is organized alphabetically by grammatical class. Nouns are further categorized by semantic class (e.g., animals, articles of clothing, places) and ordered alphabetically. Gerunds and past participles, both verb derivations, are listed as separate grammatical categories. Stable substitutes and their adult equivalent are presented in a separate appendix. Each word is followed by the FTU, AoA, stress pattern, number of syllables, number or letters and frequency of use.

The PM database

The resulting list from Manzano et al. (1997) and Piñeiro and Manzano (2000) includes a total of 1,259 different words, representing all types of word classes including 449 common nouns (36%), 385 verbs (31%), 225 proper nouns (18%), 46 stable substitutes (4%), 31 adjectives (2% %), 31 pronouns (2%), 27 adverbs (2%), 21 gerunds (2%), 9 prepositions (0.5%), 9 articles (0.5%), 9 interjections (0.5%), 8 numerals (0.5%), 5 past participles (0.5%), 2 conjunctions (0.5%), and 2 contractions (0.5%). Out of 1259, 1189 (94.5%) are open-class words (e.g., nouns, verbs, adjectives, and adverbs), and 70 (5.5%) are closed-class (e.g., articles, pronouns, and prepositions). The database also reports the data by age group: younger children were those less than 25;15 months of age; and older children were more than 25;15 months of age.

Nouns and verbs are the most common word classes in the Piñeiro and Manzano (PM) database. The combination of common nouns and proper nouns make up approximately 54% of the total word types. Approximately 31% of the word types are verbs. Of the 385 verbs, 327 (85%) were conjugated and 58 (15.06%) were produced in their infinitive form. Out of the 58 infinitive verbs, 51 were produced by the older children (88%). Younger children only produced 7 verbs in infinitive form. In Spanish, the infinitive forms of verbs have three endings: *-ar* (e.g., *jugar*, to play), *-er* (*comer*, to eat), and *-ir* (*dormir*, to sleep). The most frequent ending for the infinitive forms of verbs was *-ar* (produced 39 times or 67%).

Younger children ($\leq$25;15 months) produced 96 out of 327 conjugated verbs (29%); of those 86 (90%) were in present tense and 10 (10%) were in simple past tense (e.g., imperfect past and indefinite past). Older children ($>$ 25;15 months) produced 231 different conjugated verbs, of which 179 (77%) were in present tense, and 52 (23%) in simple past tense. The majority of conjugated verbs were produced in first person singular. There were no verbs in future or compound tenses (i.e., auxiliary verb + participles). The children used periphrasis such as "*voy a*" (I'm going to) or temporal words like *mañana* (tomorrow), *después* (afterwards), or *luego* (later) to denote future tense. Younger children produced 3 gerunds and no participles, while older children produced 18 gerunds and 5 participles.

The frequency of use for all words ranged from 1 to 444 (mean = 12; $SD = 32$). For 386 words, the frequency was 1. The frequency was greater than 100 for 31 words. Fifty two percent of high frequency words were open class words, such as nouns ($n = 6$), adverbs ($n = 6$), verbs ($n = 3$), adjectives ($n = 1$). The rest, almost half of the words (48%) were closed class items, including personal pronouns ($n = 7$), prepositions ($n = 2$), articles ($n = 3$), conjunctions ($n = 2$), and interjections ($n = 1$). Piñeiro and Manzano (2000) found that AoA was negatively correlated with frequency, suggesting that the words that children used most frequently were learned at an early age. In fact, all 31 most frequently used words were produced for the first time by children younger than 30 months of age.

Comparison between the PM database with the MBCDI in Spanish

Following the procedures in Laubscher and Light (2020), we compared the words used by children between 11 and 30 months in the PM database to the words on the MacArthur Inventarios del Desarrollo de Habilidades Comunicativas (IDHC). The IDHC is a parent report instrument developed for children aged 0;8 to 2;6 (years; months; Jackson-Maldonado et al., 1993). Like other versions of the MBCDI, the IDHC was developed to gather information about a child's early communication and language skills. The IDHC includes a word checklist of 375 nouns, grouped in semantic categories (e.g., food and drinks, people, household items, outside things, temporal, toys, etc.), 106 verbs, 63 adjectives, 43 pronouns, 15 adverbs, 13 locatives, 12 stable substitutes, 9 prepositions, 9 articles, and 6 conjunctions.

Adults filling out the IDHC are instructed to mark the words that the child understands and produces. Common immature versions of a word (*bobo* for *oso*), pronunciation errors (*tete* for *leche),* and forms substituted by the child's family (*Coca* for *refresco)* are acceptable. To observe the utilization of verb conjugations, parents are presented with only the infinitive form for each verb ending (-ar, *acabar*; -er, *comer*; and -ir, *subir*), and are asked to note their recollection of the conjugations produced by the child following any verb.

The IDHC is not a direct translation of the original English CDI. The IDHC was developed using the following primary

sources: (a) natural language samples obtained from several published and unpublished studies of Mexican Spanish and Spanish spoken in the United States (e.g., González, 1983; Jackson, 1989) and (b) corpora from language experiments carried out at the University of the Americas in Mexico City (see Jackson-Maldonado et al., 1993). The authors supplemented these sources with vocabulary lists from language or intelligence tests such as Test Vocabulario Imágenes Peabody (Dunn, Padilla, Lugo, & Dunn, 1986) and Batería de Evaluación Intelectual Kaufman (Kaufman, Kaufman, Gomez Palacio, Rangel, & Padilla, 1987).

Although there is approximately an 80% overlap between the English MBCDI and the Spanish IDHC forms, the IDHC reflects linguistic and cultural differences (Jackson-Maldonado et al., 1993). Cultural modifications include replacing common child social games, like *tortillitas* and *ojitos*, instead of patty cake and peek-a-boo, and including additional items in the "people" category such as *madrina* ("godmother") and *padrino* ("godfather"), as well as food items like *tortilla* and *chili*. Some items from the English version were discarded for being irrelevant. For example, the items "backyard", "sandbox," and "basement", were not included in the Spanish version.

There are seven different versions of the IDHC in Spanish: Argentinian, Chilean, Columbian, Cuban, European, Mexican, and Peruvian Spanish, which reflect mostly dialectal differences of the same lexical concept (e.g., to drink: *beber/tomar*; Mariscal et al., 2007). The Columbian, Mexican and European Spanish versions have been used in several normative studies, cumulatively involving thousands of infants and toddlers (e.g., Díaz et al., 2011; Jackson-Maldonado et al., 1993; Mariscal et al., 2007; Thal, Jackson-Maldonado, & Acosta, 2000). Like their English counterpart, the IDHC has been found to be reliable and valid to assess early language skills of Spanish speaking children with and without disabilities and is increasingly incorporated in early language assessment batteries. For the purpose of this study, we selected the Mexican version, as it is widely used for assessment and screening in Latin American countries and the United States.

To determine the level of overlap between the PM database and the IDHC, we developed a list which included all the different words minus proper nouns ($n = 465$) produced by children 11 to 30 months of age from the PM database. To this list, we added 732 different words and lexical variations provided in the IDHC (e.g., *botella/mamila/tetero/biberón* for bottle, which is presented as one single item in the IDHC and counted as four different words in our study). This resulted in a list of 979 different words (without duplicates). We considered a verb to be overlapping if any of the verb conjugations in the PM database matched an infinitive in the IDHC verb category. For every other category of word, we considered two words to be overlapping only when the words matched in gender, number, and aspect (e.g., *cama* vs *camita*).

Results

Out of 979 different words, 218 words were included in both the PM database and the IDHC. This reveals an overlap of 22% between both lists. The 218 words that were included in both lists are presented in Appendix A. The words are grouped by grammatical class. Approximately 14% of the overlapping words are closed-class words, whereas 86% are open-class words. Nouns account for the largest amount of overlapping words ($n = 103$; 47%), and are further classified by semantic category, including animals, people, food/drinks, household objects, body parts, toys, vehicles, places, clothing, outside objects, activities, and time. However, words from other grammatical categories are represented as well, such as verbs ($n = 47$; 22%), pronouns ($n = 20$; 9%), adverbs ($n = 13$; 6%), adjectives ($n = 11$; 5%), stable substitutes ($n = 7$; 3%), articles ($n = 5$; 2%), numerals ($n = 5$; 2%), prepositions ($n = 4$; 2%), interjections ($n = 2$; 1%), and conjunctions ($n = 1$; 1%).

Appendix A also includes the first time the word was uttered (FTU) as well as the age of acquisition (AoA) for 117 words as reported in the PM database. The FTU represents the age range in which that word was uttered for the first time by at least one child in the PM datatase. The AoA is the age range in which a given word's frequency reached 10% of its total frequency. The PM database did not report the AoA for any verbs, even for verbs with high frequency of use such as *ir*/to go, *dar*/to give, and *mirar*/to look (e.g., *voy*/I go, *dame*/give me, *mira*/look).

All but 4 of the 31 most frequently used words in the PM database were also included in the IDHC and therefore are in the Appendix A. These 27 overlapping words represent a wide range of grammatical functions and are essential to early grammaticalization in Spanish. All the pronouns, prepositions, articles and conjunctions used by children aged 30 months or younger in the PM database were also included in the IDHC, and therefore are in the Appendix A.

Furthermore, the PM database included 247 words that were not included in the IDHC. These were 121 nouns, 64 verbs, 36 stable substitutes, 7 interjections, 9 adjectives, 6 gerunds, 2 adverbs, and 2 contractions. Of those words, sixty six percent had a frequency of 10% or less ($n = 162$). The remaining words could be considered dialectical variations of the same concept (e.g., *guagua*/bus), highly contextual objects (e.g., *coco*/coconut), diminutives (*juguete-juguetico*/toy), lexical variations based on gender and number (*jarrito-jarrita*/jar), different verb conjugations, and verbs not found on the IDHC (e.g., *monto*/to ride).

The IDHC included 514 words that were not included in the PM database. These were 317 nouns, 62 verbs, 55 adjectives, 27 pronouns, 15 social routines and games, 10 adverbs, 8 locatives, 7 prepositions and articles, 6 stable substitutes, 4 conjunctions, and 3 question words. More than 80% of items in the following categories of the IDHC were not found in the PM database: Food/Drink, Household Objects, Rooms/Furniture, Clothing, Outside Objects, adjectives, and temporal concepts.

Discussion

The aim of this study was to generate a vocabulary list to support families, clinicians and AAC software developers in the selection of developmentally appropriate Spanish

vocabulary. The list is relevant for children who are in the early stages of Spanish language development. To extract the specific words in our list, we analyzed the overlap between a list of words produced by 200 Spanish speaking children between 11 and 30 months of age (Manzano et al., 1997; Piñeiro & Manzano, 2000) with a list of words taken from the IDHC (Jackson-Maldonado et al., 1993). Both of these lists have been validated and used in several normative studies (e.g., Jackson-Maldonado et al., 1993; Mariscal et al., 2007; Piñeiro et al., 1999; Thal et al., 2000).

The resulting list includes a total of 218 overlapping words. As in other languages, common nouns dominate the list (e.g., Jackson-Maldonado et al., 1993; McDonough et al., 2011). This is expected, as nouns dominate the vocabularies of Spanish-speaking young children (see McDonough et al., 2011; Quintero & Pérez, 2007). However, the 218 words also include a range of developmentally appropriate verbs, adjectives, pronouns, adverbs, articles, and prepositions, providing evidence of early use of these grammatical structures in the language of Spanish speaking children. All the articles, pronouns, prepositions, and conjunctions in the PM database were also included in the IDHC, as were 27 of its top 31 most frequently used words. The high overlap suggests that the function words in our list are representative of function words used by Spanish speaking children in the early stages of language development. In fact, the combination of article plus noun, and pronoun plus verb are among the first word combinations used by Spanish young children (Ortiz-Ciscomani & Alvarez-López, 2019; Shum, Conde, & Díaz, 1992).

The PM database did not include 514 words from the IDHC. Likewise, 247 words from the PM database were not included in the IDHC. These differences may have to do with the original purpose of both lists, the procedures used to develop them, and the criteria for inclusion of a word in each word list. The PM database was created by transcribing children's utterances during spontaneous play and conversations with peers and adults in day care centers. As such, the database included words that are relevant to that context, and many of the words had a low frequency of use.

The IDHC is an inventory of words that was developed from a combination of sources, including parental reports, language samples, and published and unpublished word lists (see Jackson-Maldonado et al., 1993 for a description). The IDHC was designed to be "a screening instrument for early Spanish acquisition … and early detection of language delay" (Jackson-Maldonado et al., 1993, p. 546). Because it is used as a tool to describe a child's early language, it includes a large number and variety of content words that are applicable in multiple contexts.

The IDHC has numerous advantages in evaluating early linguistic development in Spanish as several studies have shown (e.g., Mariscal & Gallego, 2012; Serrat, Olmo, & Sanz-Torrent, 2009). However, its limitations as a sole vocabulary selection tool for Spanish initial lexicon should also be considered. For instance, the IDHC does not fully account for the fact that Spanish is an inflectional language. Nouns and adjectives are inflected for gender, number, and aspect (e.g., diminutives). Verbs have a root with suffixes denoting person, tense, number, aspect, and mood (see Mueller Gathercole, Sebastián, & Soto, 1999 for a discussion on the acquisition of Spanish early morphology). The IDHC does not provide all possible morphological variations of a word or a verb, and does not offer any guidance on the acquisition or frequency of verb morphology (e.g., Harrison, 1992; Mueller Gathercole et al., 1999). This may lead families and professionals to select a standard form of a word or a verb inflection that may be developmentally inappropriate or even ungrammatical. For instance, the IDHC includes *yo*/I as one of the pronouns and *gustar*/to like as one of its verbs leading to the possibility of these being combined erroneously as "yo + gusto" (as observed in some AAC software in Spanish). This combination is ungrammatical for several reasons. The verb "gustar" conjugates differently from other types of verbs in Spanish. In addition, "yo + gusto" uses a combination of "personal pronoun" + verb (SV), yet Spanish speaking children seldom use this combination, as the person is always inflected in the verb. In fact, 60% of early sentences produced by children as young as 18 months of age follow VSO word order (López-Ornat, 1990, 1992).

Implications for practice

The resulting list provided in Appendix A can assist caregivers and professionals in the selection of an early Spanish lexicon. The list includes all the grammatical categories used by Spanish-speaking children in the early stages of language development, and reinforces the need for AAC practitioners to rely on natural language development and typical vocabulary acquisition patterns when selecting an initial Spanish lexicon for young children. Open class words are predominant in the list. Yet, the articles, conjunction, prepositions and pronouns included in the list are frequently used by young children, as found in natural language samples (see Piñeiro & Manzano, 2000).

As with any other languages, the approach to selecting Spanish vocabulary for children who require the use of AAC should reflect the way Spanish-speaking children with typical development learn and use their language. That requires an understanding of the underlying grammatical structure of the language, as well as the typical age and developmental path in which specific words and linguistic structures are acquired and used (see Alonso, Díez, & Fernandez, 2016; Altares, Nieva, & López-Ornat, 2010; Álvarez & Cuetos, 2007; Espinosa-Ochoa, 2019; Gascón, Soto, & Gathercole, 2004; Harrison, 1992; Perez & Navalón, 2005). The age of acquisition provided in our list can assist professionals with developmental information on the order and typical age of acquisition of each word.

The list in Appendix A includes only the standard form of a word and the infinitive form of a verb. To put this list into practice, word variations based in number (plural/singular), gender (feminine/masculine) and aspect (standard/diminutive) are suggested. Likewise, the list of verbs includes only the infinitive form. Since Spanish speaking children in early stages of language development seldom use the infinitive,

developmentally appropriate conjugations ought to be selected. These would include the present and simple past tense, the periphrastic future, the gerund and the participle (see González, 1983; Harrison, 1992; Piñeiro & Manzano, 2000).

Language development in Spanish, like in other languages, is mediated by the interdependence between lexical and grammatical domains (see Serrat et al., 2009; Serrat, Sanz-Torrent, Feijóo, Chireac, & Hilferty, 2012; Simon-Cereijido & Gutiérrez-Clellen, 2009). Our overlap analysis indicates that children as young as 30 months of age use a wide range of content and function words (words that require word combinations to become meaningful) in their speech. Function words such as articles, prepositions and pronouns are essential to early syntax and grammaticalization in Spanish, and they are present in the early vocabulary of Spanish speaking children with typical development. Without these words, children would be extremely limited in their capacity to communicate. However, an initial lexicon in Spanish should also include content and child-specific words that reflect the unique interest and experiences of the child, including personally relevant proper and common nouns, social and routine words, sound effects, stable substitutes, diminutives, and other early emerging concepts that allow children to communicate successfully before they are able to produce more grammatically complex utterances (Laubscher & Light, 2020).

Although helpful, published word lists can never provide the complete solution to vocabulary selection in any language, and should always be used in conjunction with other vocabulary selection strategies (Bean et al., 2019; Trembath et al., 2007). Caregiver and informant interviews, peer observations, natural language samples, and developmental language guides can help ensure that the vocabulary selected is relevant and tailored to each child's needs and stage of language development, personality, interests, and contexts in which they interact (Trembath et al., 2007).

Limitations and future research direction

The results of the present study should be interpreted with caution because of a number of limitations. First, the study relied only on three published word lists to create a proposed initial Spanish lexicon. It is possible that there are more published and unpublished sources of early Spanish vocabulary that were not included in our review. Second, the lists we used to generate our final list were derived from studies that included only children with typical development. Future studies are needed to investigate vocabulary use in Spanish-speaking children with disabilities in the early stages of language development.

Third, our list (Appendix A) is being proposed as a pool from which to select an initial lexicon in Spanish. Further examination is required to determine the vocabulary that would be appropriate and useful for individuals in more advanced levels of language development. Future research studies should also examine whether variables such as FTU and AoA as defined by Piñeiro and Manzano (2000) hold promise for describing language acquisition patterns among individuals who use AAC.

Last, our study is an initial and limited step toward assisting families and professionals with Spanish vocabulary selection for children who use AAC. To establish the robustness of our list, future studies should examine the early vocabularies of children across different dialectal variations of Spanish, as well as compare the expressive vocabulary of monolingual and bilingual Spanish speaking children (see Mancilla-Martinez, Pan, & Vagh, 2011), and of children with and without typical development.

Conclusion

This paper represents the first work to generate a vocabulary list that can be used in the selection of a Spanish initial lexicon for children in need of AAC. It is proposed as a starting point from which to spark attention in the AAC field to the study of vocabulary selection in Spanish and should not be regarded as a finite set of words that would cover all the communicative needs of a child in need of AAC. Rather, these words can be used to select possible targets for vocabulary selection for users of AAC in the early stages of Spanish development. Our list should be supplemented with other tools that would yield vocabulary items that are representative of the child's interests, family, and culture (Mngomezulu et al., 2019; Soto & Yu, 2014; Tonsing & Soto, 2020). Similarly to any child using AAC, children who are learning Spanish also need access to individualized vocabularies that are reflective of their interests, personalities, contexts in which they interact, and dialects preferred by their parents or significant others (Trembath et al., 2007).

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Appendix A

List of Overlapping Words Between the Piñeiro and Manzano (PM) Database and the Inventorios del Desarrollo de Habilidades Comunicativas (IDHC)

Word	FTU	AoA
NOUNS		
Animals		
ardilla	22;16-25;15	
bicho	20;16-22;15	
caballo	14;16-17;15	20;16-22;15
conejo	20;16-22;15	20;16-22;15
gallina	20;16-22;15	
gato	14;16-17;15	17;16-20;15
jirafa	26;4-29;16	
león	22;16-25;15	
mariposa	17;16-20;15	
mono	20;16-22;15	22;16-25;15
osito	20;16-22;15	22;16-25;15
oso	22;16-25;15	38;4-41;23
pájaro	22;16-25;15	
pato	14;16-17;15	20;16-22;15
perro	14;16-17;15	22;16-25;15
pescado	20;16-22;15	34;7-37;21
pingüino	22;16-25;15	
pollito	20;16-22;15	20;16-22;15
pollo	22;16-25;15	
rana	20;16-22;15	22;16-25;15
ratón	22;16-25;15	
vaca	20;16-22;15	20;16-22;15
People		
abuela	22;16-25;15	34;7-37;21
abuelo	22;16-25;15	34;7-37;21
amigo	26;4-29;16	
hermana	26;4-29;16	
hermano	26;4-29;16	
maestra	26;4-29;16	
mamá*	14;16-17;15	22;16-25;15
niña	22;16-25;15	26;4-29;16
niño	17;16-20;15	22;16-25;15
papá*	11;16-14;15	22;16-25;15
payaso	22;16-25;15	22;16-25;15
policía	26;4-29;16	
prima	26;4-29;16	26;4-29;16
primo	26;4-29;16	26;4-29;16
tía	14;16-17;15	14;16-17;15
Food and Drink		
agua*	11;16-14;15	22;16-25;15
azúcar	26;4-29;16	
calabaza	26;4-29;16	
chocolate	26;4-29;16	
comida	22;16-25;15	30;4-33;22
dulce	26;4-29;16	
galleta	22;16-25;15	30;4-33;22
helado	26;4-29;16	26;4-29;16
hielo	20;16-22;15	
jugo	22;16-25;15	30;4-33;22
leche	20;16-22;15	22;16-25;15
naranja	22;16-25;15	
papa	20;16-22;15	22;16-25;15
Household Objects		
cuchara	22;16-25;15	22;16-25;15
cuna	26;4-29;16	
escalera	22;16-25;15	
escoba	26;4-29;16	
espejo	22;16-25;15	
luz	20;16-22;15	
mesa	22;16-25;15	
peine	26;4-29;16	
plato	22;16-25;15	22;16-25;15
puerta	20;16-22;15	22;16-25;15
silla	17;16-20;15	22;16-25;15
teléfono	20;16-22;15	22;16-25;15
vaso	22;16-25;15	

(*continued*)

Continued.

Word	FTU	AoA
Body Parts		
boca	20;16-22;15	
cara	26;4-29;16	
mano	22;16-25;15	22;16-25;15
nariz	17;16-20;15	17;16-20;15
ojo	26;4-29;16	30;4-33;22
oreja	26;4-29;16	
pelo	14;16-17;15	22;16-25;15
pie	22;16-25;15	30;4-33;22
teta	20;16-22;15	
Toys		
carriola	22;16-25;15	
colores	22;16-25;15	38;4-41;23
globo	26;4-29;16	
jugete	17;16-20;15	26;4-29;16
lápiz	22;16-25;15	
muñeca	20;16-22;15	22;16-25;15
pala	22;16-25;15	
papel	17;16-20;15	
pelota*	17;16-20;15	22;16-25;15
Vehicles		
avión	17;16-20;15	20;16-22;15
barco	22;16-25;15	30;4-33;22
camión*	20;16-22;15	22;16-25;15
carro	22;16-25;15	22;16-25;15
coche	22;16-25;15	34;7-37;21
moto	22;16-25;15	22;16-25;15
tractor	22;16-25;15	
tren	20;16-22;15	20;16-22;15
Places		
calle	22;16-25;15	34;7-37;21
casa	22;16-25;15	30;4-33;22
cocina	22;16-25;15	
parque	22;16-25;15	22;16-25;15
playa	22;16-25;15	26;4-29;16
Clothing		
reloj	11;16-14;15	20;16-22;15
sombrero	20;16-22;15	
zapato	17;16-20;15	20;16-22;15
Outside Objects		
estrella	26;4-29;16	
flor	20;16-22;15	34;7-37;21
sol	20;16-22;15	26;4-29;16
Activities		
besito	26;4-29;16	26;4-29;16
fiesta	22;16-25;15	38;4-41;23
Time		
tarde	22;16-25;15	
VERBS		
abrir	22;16-25;15	
acabar	22;16-25;15	
bailar	26;4-29;16	
buscar	22;16-25;15	
caer(se)	22;16-25;15	
caminar	22;16-25;15	
cantar	22;16-25;15	
cerrar	22;16-25;15	
cocinar	26;4-29;16	
comer(se)	22;16-25;15	
comprar	22;16-25;15	
correr	22;16-25;15	
dar*	11;16-14;15	
dormir(se)	22;16-25;15	
enseñar	22;16-25;15	
escuchar	14;16-17;15	
esperar(se)	22;16-25;15	
estar	20;16-22;15	
gustar	26;4-29;16	
haber	22;16-25;15	
hablar	26;4-29;16	
hacer	22;16-25;15	
ir(se)	22;16-25;15	
jugar	20;16-22;15	

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Word	FTU	AoA
llevar(se)	26;4-29;16	
llorar	17;16-20;15	
mirar*	11;16-14;15	
morder	26;4-29;16	
oír	20;16-22;15	
parar(se)	22;16-25;15	
peinar(se)	26;4-29;16	
perder	20;16-22;15	
poner(se)	22;16-25;15	
querer	22;16-25;15	
quitar	17;16-20;15	
romper	22;16-25;15	
saber	22;16-25;15	
salir(se)	22;16-25;15	
saltar	26;4-29;16	
sentar(se)	22;16-25;15	
ser	20;16-22;15	
tener	22;16-25;15	
tirar	20;16-22;15	
tocar	22;16-25;15	
tomar(se)*	26;4-29;16	
venir(se)	17;16-20;15	
ver(se)	17;16-20;15	
PRONOUNS		
aquella	22;16-25;15	42;6-45;21
dónde	20;16-22;15	22;16-25;15
esa	22;16-25;15	26;4-29;16
ese	20;16-22;15	22;16-25;15
eso	22;16-25;15	34;7-37;21
esta	22;16-25;15	26;4-29;16
este	17;16-20;15	22;16-25;15
esto*	17;16-20;15	22;16-25;15
le	22;16-25;15	22;16-25;15
me*	22;16-25;15	34;7-37;21
mi*	11;16-14;15	34;7-37;21
mía*	17;16-20;15	20;16-22;15
mío*	14;16-17;15	22;16-25;15
por qué	22;16-25;15	38;4-41;23
que	22;16-25;15	30;4-33;22
qué	17;16-20;15	22;16-25;15
quién	22;16-25;15	22;16-25;15
se*	20;16-22;15	22;16-25;15
tuyo	22;16-25;15	34;7-37;21
yo*	17;16-20;15	34;7-37;21
ADVERBS		
abajo	26;4-29;16	
ahí*	11;16-14;15	22;16-25;15
allá*	11;16-14;15	38;4-41;23
allí	22;16-25;15	22;16-25;15
aquí*	17;16-20;15	20;16-22;15
arriba	22;16-25;15	30;4-33;22
más	17;16-20;15	20;16-22;15
mucho*	22;16-25;15	26;4-29;16
no*	11;16-14;15	22;16-25;15
otra	22;16-25;15	
otro	17;16-20;15	20;16-22;15
sí	17;16-20;15	20;16-22;15
ya	11;16-14;15	14;16-17;15
ADJECTIVES		
amarillo	20;16-22;15	30;4-33;22
azul	20;16-22;15	22;16-25;15
blanco	22;16-25;15	
chico	20;16-22;15	
feo	22;16-25;15	26;4-29;16
fria	22;16-25;15	
grande*	22;16-25;15	30;4-33;22
limpio	26;4-29;16	
lindo	20;16-22;15	22;16-25;15
rojo	20;16-22;15	26;4-29;16
verde	22;16-25;15	38;4-41;23
STABLE SUBSTITUTES		
cua cua (pato)	14;16-17;15	
gua gua (perro)	14;16-17;15	17;16-20;15
kikiriki (gallo)	17;16-20;15	
miau (gato)	17;16-20;15	20;16-22;15
mu (vaca)	14;16-17;15	
pío pío (pollo)	22;16-25;15	22;16-25;15
pipi (carro)	14;16-17;15	17;16-20;15
ARTICLES		
el*	14;16-17;15	22;16-25;15
la*	11;16-14;15	22;16-25;15
lo	22;16-25;15	26;4-29;16
un*	14;16-17;15	22;16-25;15
una	26;4-29;16	30;4-33;22
NUMERALS		
cinco	20;16-22;15	38;4-41;23
cuatro	20;16-22;15	38;4-41;23
dos	20;16-22;15	26;4-29;16
tres	20;16-22;15	34;7-37;21
uno	20;16-22;15	22;16-25;15
PREPOSITIONS		
a*	17;16-20;15	22;16-25;15
de*	22;16-25;15	34;7-37;21
en	22;16-25;15	30;4-33;22
para*	22;16-25;15	22;16-25;15
INTERJECTIONS		
adiós	20;16-22;15	
gracias	22;16-25;15	
CONJUNCTIONS		
y*	20;16-22;15	30;4-33;22

Note. FTU = First Time Uttered or age range in which a word was uttered for the first time by at least one child; AoA = Age of Acquisition or the age range in which a given word's frequency reached 10% of its total frequency. The PM Database did not report AoA on any verb or any word whose frequency was lower than 10.

* Reflects a high frequency word (>100) according to the PM database.

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